

Prostart - Presentation Script

1. PROJECT INTRODUCTION - Slide 2

Hi everyone! We are Arianna, Duarte, Lorenzo, Loke and Ninad, and our project is called Prostart. [Play the video.] In sprinting, races are often won by a few thousandths of a second, and part of that margin comes from the reaction to the gun. But measuring reaction time with millisecond accuracy needs expensive systems, and in practice only federations and large clubs own them. Our idea is simple, we designed a small, low-cost box that mounts on any existing starting block. The box gives the start signal, "on your marks", "set", "go", and measures the athlete's reaction time with an accelerometer. A phone at the finish line records the finish, like a photo finish. It's built specifically for club coaches and solo athletes who currently have to rely on basic stopwatches. This technology effectively bridges the gap between professional timing and everyday training.

2. PROBLEM STATEMENT - Slide 3

In competitive sprinting, measuring block-start reaction time with millisecond accuracy is a luxury non-elite athletes and coaches can rarely afford. Yet, fractions of a second change everything, let's think for example at Noah Lyles, who won the Paris Olympic 100-metre final by a mere five thousandths of a second, a margin where a fast and well-measured start plays a critical role. Any reaction under 0.100 seconds is flagged as a false start, and elite sprinters sit only slightly above that mark.. So what do most clubs use today? Hand timing. But a manual stopwatch introduces an error of about 0.24 seconds compared to electronic systems, rendering times completely unreliable for the tiny gains an athlete trains for. Other solutions exist. Photocells start timing when the body crosses a beam, so they miss the reaction. FinishLynx ReactTime measures reaction time with a threshold on acceleration, but even the training version for one block costs about 1,600 dollars. At the Olympics, Omega uses force sensors built into the blocks. So there is a clear gap: a low-cost, portable system with a documented timing error.

3. TECHNICAL APPROACH - Slide 4

Our system has two parts: a start unit on the block, and a phone at the finish line. The start unit is built on a compact board, the Seeed XIAO nRF52840 Sense, which has an accelerometer on it. It reads the accelerometer 865 times per second, giving us one sample every 1.2 milliseconds. To find the exact moment the athlete starts to push, we use two steps. First, an STA/LTA trigger decides that a movement happened. Then an AIC picker looks around that moment and finds the sample where the signal changes from quiet to active. This second step does not depend on any threshold. The start unit

sends the start time by wire to an ESP32, which sends it by radio, with ESP-NOW, to a second ESP32 connected to the phone. We write the firmware in Arduino, the app in Flutter, and the analysis in Python.

4. USER INTERACTION - Slide 5

Now, how does an athlete use it? First, the coach or athlete mounts and securely fixes the box onto the back of the starting block. Then they press one button. The box plays three beeps: "on your marks", "set" and "go". After "set", the box waits until the athlete is completely still, and then it fires "go" at a random time, between 0.7 and 1.5 seconds later. When the athlete pushes off, the box measures the reaction time. If the athlete moves less than 100 milliseconds after "go", it is a false start. At the finish line, the phone app films all the time. When it receives the start time from the block, AI on the phone finds the moment the athlete's torso crosses the line. The finish time is simply the finish moment minus the start moment. Here you can see the design of our app.

5. RESULTS - Slide 6-7

Here are our results. This is our first prototype on a breadboard, mounted on the back of a real block. So far, we have built and tested the start unit. Five sprinters performed 79 attempts over four sessions: 72 valid starts, with a median reaction time of 154 milliseconds, and 7 false starts. [Next slide.] Our main result is shown in this graph. We swept the detection threshold from 10 to 50 milli-g. With a simple threshold—the red line—the reported reaction time drifts, with a median shift of 9.3 milliseconds. With our two-step detector, the blue line remains completely stable: in 70 of 79 starts the result did not change at all, and in 74 it stayed within a single sample. We estimate the random error at 1 to 2 milliseconds. However, a stable number is not yet a guaranteed correct number, so our next step will be to validate it against a mechanical ground truth.

6. NEXT FUTURE- Slide 8-9

Looking ahead, our next steps are structured into four main phases. First, we will validate our reaction time against a mechanical ground truth, because a stable number is not yet a guaranteed correct number. Second, we will close the wireless link: after exploring options like Wi-Fi, Zigbee, and Bluetooth and noting that our initial Wi-Fi tests with the ESP32-C3 only reached 60 to 70 metres, whereas a full track requires 150 to 200, we are implementing a robust long-range ESP-NOW connection from the start unit to the finish line. Third, we will build the photo finish app in which on-device AI detects the athlete's torso crossing the line. Finally, we will validate the photo finish against a 1000 fps camera. While full track validation and the photo-finish integration are still ahead, PROSTART is proving that Olympic-level timing accuracy can be accessible to everyone. Thank you!